(Please write your Enrollment Number) Enrollment No. 035

End-Term Examination (Regular & Reappear)
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech (CSE/IT/ CSE-AI/ AI&ML), Semester: 3rd Sem.
(November-December, 2023)

Subject Code: BCS 203	Subject: Discrete Structures
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 — **2.5*8=20**

a) Let p, q r, s represents the following propositions
- p: $x \in \{8, 9, 10, 11, 12\}$
- q: x is a composite number
- r: x is a perfect square.
- s: x is a prime number.

The integer $x \geq 2$ which satisfies $\neg((p \rightarrow q) \wedge (\neg r \vee \neg s))$ is/are _____.

b) Translate the following proposition into logical formula :
i) "Not all that glitters is gold" where Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold.
ii) "Some boys in the class are taller than all the girls", where Predicate boy(x) is true if x boy and predicate girl(x) is true if x is girl, and taller(x,y) is true if x is taller than y.

c) Write the Hasse diagram of power set P(A) over the relation subset. Where A = {a, b, c, d}.

d) Define the partial order relation with example.

e) Find the total number of elements between 10 and 1000, which are divisible by either 3 or 5.

f) Identify whether the given graph is planar with explanation?

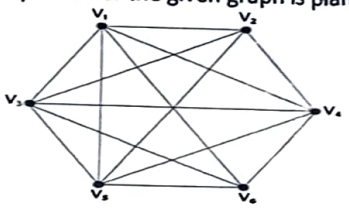
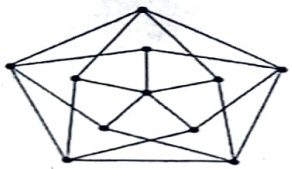
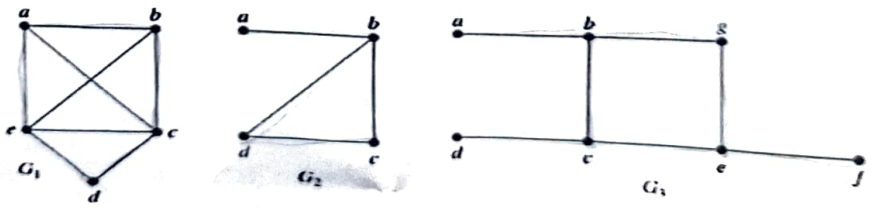


Fig: G_1

g) Find the chromatic number of the following graph with explanation.

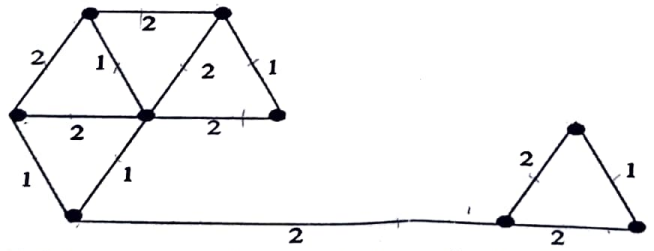
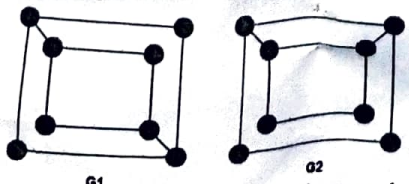


h) Examine and justify your answer, whether the following graphs are Hamiltonian path or circui



UNIT-I

Q2 a) Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed." **(5+5)**

	b) Prove that by mathematical induction For all $n \geq 1$, prove that, $1^2 + 2^2 + 3^2 + \ldots + n^2 = \{n(n+1)(2n+1)\}/6$	
Q3	a) Translate the following statements: i) $\forall x \forall y(((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$ ii) $\forall x \forall y(((x < 0) \wedge (y < 0)) \rightarrow (xy > 0))$ b) Identify the following Boolean expressions which is/are NOT tautology? A. $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$ B. $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$ C. $(a \wedge b \wedge c) \rightarrow (c \vee a)$ D. $a \rightarrow (b \rightarrow a)$	(5+5)
	UNIT-II	
Q4	a) Define antisymmetric relation and find out the total number of antisymmetric relation possible with a set of n natural numbers. b) Prove that set of integer Z is countable infinite set.	(10)
Q5	a) Let A = R- {3} and B = R – {1}. Consider the function f: A →B defined by f (x) = (x- 2)/(x -3). Is f one-one and onto? Justify your answer. b) What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?.	(5+5)
	UNIT-III	
Q6	a) Prove that a given set $Z_5 = (\{0, 1, 2, 3, 4\}, \times_5, +_5)$ is a field. b) Prove that the order of the subgroup H is the divisor of the order of the group G.	(5+5)
Q7	a) Let G be the set of all positive rational numbers and * be the binary operation on G defined by $a * b = \frac{ab}{7}$ for all $a, b \in G$. Prove that (G, *) is an abelian group. b) Prove that every group of prime order is cyclic.	(5+5)
	UNIT-IV	
Q8	a) The number of distinct minimum spanning trees for the weighted graph below is _____, explained in detail.  b) Let G be a connected planar simple graph with V vertices, E edges and R regions, then prove that R = E-V + 2.	(5+5)
Q9	a) Identify whether the following graphs, G1 and G2, are isomorphic.  b) Given preorder and postorder traversal sequences of a binary search tree, can we always construct a unique binary tree? Justify your answer with an example.	(5+5)

7th Dec
S2
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